

1) What is the effect of the catalyst on the of activation energy?

However, if a catalyst is added to the reaction, the activation energy is lowered because a lower energy transition state is formed. Enzymes can be thought of as biological catalysts that lower activation energy.

- A catalyst provides a reaction route with a lower activation energy.
- Suppose that the catalyzed activation energy is to 25 kJ mol⁻¹.
- The calculation is represented at 293 K:

$$e^{-\frac{E_A}{RT}} = e^{-\frac{25000}{8.31 \times 293}}$$
$$= 3.47 \times 10^{-5}$$

- Compared with the corresponding value for activation energy of 50 kJ mol⁻¹, there is a significant increase in the fraction of molecules able to react.
- There are almost 30,000 times more molecules which can react in the presence of the catalyst compared to having no catalyst (using our assumptions about the activation energies).
- These calculations can also be done in reverse: given the rate of reaction or rate constants at several temperatures, the activation energy can be calculated.

2) What is pseudo first order reaction?

Under certain conditions, the second order kinetics can be well approximated (might appear an in experiment) as first order kinetics. This is called pseudo first order reaction. These Pseudo first order reactions greatly simplify quantifying the reaction dynamics. The functional form of the decay kinetics is similar to the first order kinetics and the system is said to operate under pseudo first order kinetics. In a pseudo first order reaction, we can manipulate the initial concentrations of the reactants.

3) Define solubility?

The solubility of a solution is a measure of how much of the solute can be dissolved into the solvent. The solution reaches a point called the saturation point when no more solute will be accepted by the solvent. Any further addition of solute will result in solid solute mixed in with the saturated solution. Each solvent and solute pair has a characteristic solubility at a given temperature. Usually as you increase the temperature, an increased amount of solute will be able to dissolve.

4) What are elementary steps for the reaction mechanism?

Based on numbers of molecules involved in the elementary step, there are three kinds of elementary steps:

1. Unimolecular step
2. Bimolecular step
3. Trimolecular step

1. Unimolecular Step

When a molecule or ion decomposes by itself, such an elementary step is called a unimolecular step (or process). A unimolecular step is always a first order reaction. The following examples are given to illustrate this point:



Or in general

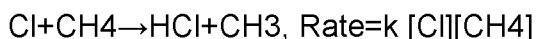
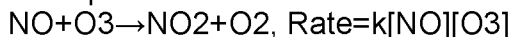


A* represents an excited molecule.

2. Bimolecular Step

A bimolecular process involves two reacting molecules or ions. The rates for these steps are 2nd order, and some examples are given to illustrate how you should give the rate expression. The simulation illustrates a bimolecular process.

Examples:

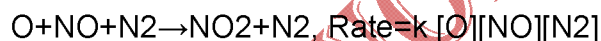
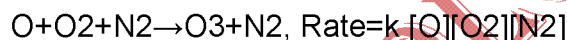


In general:



3. Trimolecular Step (process)

A trimolecular process involves the collision of three molecules. For example:



The N₂ molecules in the above trimolecular elementary steps are involved with energy transfer. They cannot be canceled. They are written in the equation to give an expression for the Rates.

In general, trimolecular steps may be,



Three molecules colliding at an instant is rare, but occasionally these are some of the ways reactions take place.

5) Write Properties of colloids?

- The particles of dispersant are between about 500 nm to 1 nm in diameter.
- The mixture does not separate on standing in a standard gravity condition. (One 'g'.)
- The mixture does not separate by common fiber filter, but might be filterable by materials with a smaller mesh.
- The mixture is not necessarily completely homogeneous, but usually close to being so.

- The mixture may appear cloudy or almost totally transparent, but if you shine a light beam through it, the pathway of the light is visible from any angle. This scattering of light is called the Tyndall effect
- There usually is not a definite, sharp saturation point at which no more dispersant can be taken by the dispersing agent.
- The dispersant can be coagulated, or separated by clumping the dispersant particles with heat or an increase in the concentration of ionic particles in solution into the mixture.
- There is usually only small effect of any of the colligative properties due to the dispersant.

6) What is mechanism of stopped flow method of concentrations?

- ☐ ☐ This technique involves two reactants held in separate reservoirs that are prevented from freely flowing by syringe pumps.
- ☐ ☐ The reaction is initiated by depressing the reactant syringes, and thus releasing the reactants into the connecting "mixing chamber" where the solutions are mixed.
- ☐ ☐ The reaction is monitored by observing the change in absorbance of the reaction solution as a function of time.
- ☐ ☐ As the reaction progresses it fills the "stop syringe" which then expands until it hits a block at the point when the reaction has reached a continuous flow rate, thereby stopping the flow and the reaction, and thus allowing the researcher to calculate the exact initial rate of reaction.
- ☐ ☐ The Stopped Flow technique works because within milliseconds of combining the two reactants the absorbance can be read.
- ☐ ☐ In addition, the stop syringe assures for a steady rate of flow past the spectrophotometer so that reactants are being added to solution and forming products at a consistent rate.

7) What are Azeotropes?

An azeotrope is a mixture that exhibits the same concentration in the vapor phase and the liquid phase. Azeotropes are a mixture of at least two different liquids. This is in contrast to ideal solutions with one component typically more volatile than the other; this is the reason we use distillation to separate materials. If the mixture forms an azeotrope, the vapor and the liquid concentrations are the same, which preventing separation via this approach. Their mixture can either have a higher boiling point than either of the components or they can have a lower boiling point. Azeotropes occur when fraction of the liquids cannot be altered by distillation. Typically when dealing with mixtures, components can be extracted out of solutions by means of Fractional Distillation, or essentially repeated distillation in stages (hence the idea of 'fractional'). The more volatile component tends to vaporize and is collected separately while the least volatile component remains in the distillation container and ultimately, the result is two pure, separate solutions.

8) What are the limitations of thermodynamics?

Thermodynamics is a powerful approach toward understanding chemical reactions, but only provides part of the picture.

Specifically:

- Thermodynamics only points the way

- Thermodynamics says nothing about how long it takes to get there
- The stoichiometric equation for the reaction says nothing about its mechanism

Thermodynamics only points the way

Chemical change is driven by the tendency of atoms and molecules to rearrange themselves in a way that results in the maximum possible dispersion of thermal energy into the world. The observable quantity that measures this spreading and sharing of energy is the free energy of the system. As a chemical change takes place, the quantities of reactants and products change in a way that leads to a more negative free energy. When the free energy reaches its minimum possible value, there is no more net change and the system is said to be in equilibrium. The beauty of thermodynamics is that it enables us to unfailingly predict the net direction of a reaction and the composition of the equilibrium state even without conducting the experiment; the standard free energies of the reactants and products, which can be independently measured or obtained from tables, are all we need.

The stoichiometric equation for the reaction says nothing about its mechanism:

The term "mechanism" refers to, "who does what to whom". Think of a reaction mechanism as something that goes on in a "black box" that joins reactants to products. □The inner workings of the black box are ordinarily hidden from researchers, are highly unpredictable, and can only be inferred by indirect means. The stoichiometric equation for the reaction says nothing about its mechanism!

Thermodynamics says nothing about how long it takes to get there:

It is worth noting that the concept of "time" plays no role whatsoever in thermodynamics. But kinetics is all about time. The "speed" of a reaction (how long it takes to reach equilibrium) bears no relation at all to how spontaneous it is (as given by the sign and value of ΔG°) or whether it is exothermic or endothermic (given by the sign of ΔH°). Moreover, there is no way that reaction rates can be predicted in advance; each reaction must be studied individually.

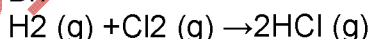
Examples:

Consider, for example, the gas-phase formation reactions of the hydrogen halides from the elements. The thermodynamics of these reactions are all similar (they are all highly exothermic), but their dynamics (their kinetics and mechanisms) could not be more different. $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightarrow 2\text{HI}(\text{g})$

Careful experiments, carried out over many years, are consistent with the simplest imaginable mechanism: a collision between the two reactant molecules results in a rearrangement of the bonds.



One might be tempted to suppose that this would proceed in a similar way, but experiments reveal that the mechanism of this reaction is far more complex. The reaction takes place in a succession of steps, some of which involve atomic H and Br.



The mechanism of this reaction is different again. Although the first two reactions reach equilibrium in minutes to an hour or so at temperatures of 300 to 600 K, a mixture of hydrogen and chlorine will not react at all in the dark, but if you shine a light on the mixture, it goes off with a bang as the instantaneous reaction releases

heat and expands the gas explosively. What is particularly noteworthy is that these striking differences cannot be reliably predicted from theory; they were revealed only by experimentation.

9) Define the half life of reaction, Discuss half life of 0, 1st and 2nd order reactions?

Half life of reaction

The half life of a reaction, $t_{1/2}$, is the amount of time needed for a reactant concentration to decrease by half compared to its initial concentration.

In zero order reactions/kinetics, the rate of a reaction does not depend on the substrate concentration. In other words, saturating the amount of substrate does not speed up the rate of the reaction. Below is a graph of time (t) vs. concentration [A] in a zero order reaction, several observation can be made: the slope of this plot is a straight line with negative slope equal negative k, the half life of zero order reaction decrease as the concentration decrease.

We learn that the zero order kinetic rate law is as followed, where [A] is the current concentration, $[A]_0$ is the initial concentration, and k is the reaction constant and t is time:

$$[A] = [A]_0 - kt$$

We would end up with a formula as such depict how long it takes for the initial concentration to dwindle by half:

$$t_{1/2} = \frac{[A]_0}{2k}$$

The $t_{1/2}$ formula for a zero order reaction suggests the half life depends on the amount of initial concentration and rate constant.

Half Life of 1st Order Reactions

In First order reactions, the graph represents the half life is different from zero order reaction in a way that the slope continually decreases as time progresses until it reaches zero. We can also easily see that the length of half life will be constant, independent of concentration. For example, it takes the same amount of time for the concentration to decrease from one point to another point. In order to solve the half life of first order reactions, we recall that the rate law of a first order reaction was:

$$[A] = [A]_0 e^{-kt}$$

To find the half life we need to isolate t and substitute [A] with $[A]_0/2$, we end up with an equation looking like this:

$$t_{1/2} = \frac{\ln 2}{k} \approx \frac{0.693}{k}$$

The formula for $t_{1/2}$ shows that for first order reactions, the half life depends solely on the reaction rate constant, k. We can visually see this on the graph for first order reactions when we note that the amount of time between one half life and the next are the same. Another way to see it is that the half life of a first order reaction is independent of its initial concentration.

Half Life of 2nd order Reactions

Half life of second order reactions shows concentration [A] vs. time (t), which is similar to first order plots in that their slopes decrease to zero with time. However, second order reactions decrease at a much faster rate as the graph shows. We can

also note that the length of half life increase while the concentration of substrate constantly decreases, unlike zero and first order reaction.

In order to solve for half life of second order reactions we need to remember that the rate law of a second order reaction is:

$$\frac{1}{A} = \frac{1}{[A]_0} + kt$$

As in zero and first order reactions, we need to isolate T on its own:

$$t_{1/2} = \frac{1}{k[A]_0}$$

This replacement represents half the initial concentration at time, t (depicted as t_{1/2}). We then insert the variables into the formula and solve for t_{1/2}. The formula for t_{1/2} shows that for second order reactions, the half life only depends on the initial concentration and the rate constant.

10) What are factors that affect the reaction rate explain?

Factors That Affect Reaction Rates:

- Although a balanced chemical equation for a reaction describes the quantitative relationships between the amounts of reactants present and the amounts of products that can be formed, it gives us no information about whether or how fast a given reaction will occur.
- This information is obtained by studying the chemical kinetics of a reaction, which depend on various factors: reactant concentrations, temperature, physical states and surface areas of reactants, and solvent and catalyst properties if either are present.
- By studying the kinetics of a reaction, chemists gain insights into how to control reaction conditions to achieve a desired outcome.

Concentration Effects

- ☐ ☐ Two substances cannot possibly react with each other unless their constituent particles (molecules, atoms, or ions) come into contact.
- ☐ ☐ If there is no contact the reaction will be zero order.
- ☐ ☐ Conversely, the more reactant particles that collide per unit time, the more often a reaction between them can occur.
- ☐ ☐ Consequently, the reaction rate usually increases as the concentration of the reactants increases.

Temperature Effects

- ☐ ☐ Increasing the temperature of a system increases the average kinetic energy of its constituent particles.
- ☐ ☐ As the average kinetic energy increases, the particles move faster and collide more frequently per unit time and possess greater energy when they collide. Both of these factors increases the rate of reaction.
- ☐ ☐ Hence the reaction rate of virtually all reactions increases with increasing temperature.

□□Conversely, the reaction rate of virtually all reactions decreases with decreasing temperature.

□□For example, refrigeration retards the rate of growth of bacteria in foods by decreasing the reaction rates of biochemical reactions that enable bacteria to reproduce.

Phase and Surface Area Effects:

- When two reactants are in the same fluid phase, their particles collide more frequently than when one or both reactants are solids (or when they are in different fluids that do not mix).
- If the reactants are uniformly dispersed in a single homogeneous solution, then the number of collisions per unit time depends on concentration and temperature, as we have just seen.
- If the reaction is heterogeneous, however, the reactants are in two different phases, and collisions between the reactants can occur only at interfaces between phases.
- The number of collisions between reactants per unit time is substantially reduced relative to the homogeneous case, and, hence, so is the reaction rate.
- The reaction rate of a heterogeneous reaction depends on the surface area of the more condensed phase
- Automobile engines use surface area effects to increase reaction rates.
- Gasoline is injected into each cylinder, where it combusts on ignition by a spark from the spark plug.
- The gasoline is injected in the form of microscopic droplets because in that form it has a much larger surface area and can burn much more rapidly than if it were fed into the cylinder as a stream.
- Similarly, a pile of finely divided flour burns slowly (or not at all), but spraying finely divided flour into a flame produces a vigorous reaction.

Solvent Effects:

The nature of the solvent also reaction rates of the solute particles.

□□For example, a sodium acetate solution reacts with methyl iodide in an exchange reaction to give methyl acetate and sodium iodide.

□□This reaction occurs 10 million times more rapidly in the organic solvent dimethylformamide [(CH₃)₂NCHO] than it does in methanol (CH₃OH).

□□Although both are organic solvents with similar dielectric constants (36.7 for DMF versus 32.6 for methanol), methanol is able to hydrogen bond with acetate ions, whereas DMF cannot.

□□Hydrogen bonding reduced the activity of the oxygen atoms in the acetate ion.

□□Solvent viscosity is also important in determining rates of reactions.

□□In highly viscous solvents, dissolved particles diffuse much more slowly than in less viscous solvents and can collide less frequently per unit time.

□□Thus the reaction rates of most reactions decrease rapidly with increasing solvent viscosity.

Catalyst Effects

□□A catalyst is a substance that participates in a chemical reaction and increases the reaction rate without undergoing a net chemical change itself.

□□ Consider, for example, the decomposition of hydrogen peroxide in the presence and absence of different catalysts.

□□ Because most catalysts are highly selective, they often determine the product of a reaction by accelerating only one of several possible reactions that could occur. Most of the bulk chemicals produced in industry are formed with catalyzed reactions.

□□ Recent estimates indicate that about 30% of the gross national product of the United States and other industrialized nations relies either directly or indirectly on the use of catalysts.

11) Calculate of Ea using Arrhenius Equation?

As temperature increases, gas molecule velocity also increases (according to the kinetic theory of gas).

This is also true for liquid and solid substances. The (translational) kinetic energy of a molecule is proportional to the velocity of the molecules ($KE = 1/2 mv^2$).

Therefore, when temperature increases, KE also increases; as temperature increases, more molecules have higher KE, and thus the fraction of molecules that have high enough KE to overcome the energy barrier also increases.

The fraction of molecules with energy equal to or greater than E_a is given by the exponential term

$$e^{\frac{-E_a}{RT}}$$

In the Arrhenius equation:

$$K = Ae^{\frac{-E_a}{RT}}$$

k is the rate constant

E_a is the activation energy

R is the gas constant

T is temperature in Kelvin

A is frequency factor constant or also known as pre-exponential factor or Arrhenius factor. It indicates the rate of collision and the fraction of collisions with the proper orientation for the reaction to occur.

Taking the natural log of both sides of Equation yields the following:

$$\ln k = \ln K = \ln Ae^{\frac{-E_a}{RT}}$$

Graphing $\ln k$ vs. $1/T$ yields a straight line with a slope of $-E_a/R$ and a y-intercept of $\ln A$.

As indicated in next Figure, the reaction with a higher E_a has a steeper slope; the reaction rate is thus very sensitive to temperature change. In contrast, the reaction with a lower E_a is less sensitive to a temperature change. Because radicals are extremely reactive, E_a for a radical reaction is 0; an Arrhenius plot of a radical reaction has no slope and is independent of temperature.

12) Define Solution?

A solution is a mixture of materials, one of which is usually a fluid. A fluid is a material that flows, such as a liquid or a gas. The fluid of a solution is usually the solvent. The material other than the solvent is the solute. We say that we dissolve

the solute into the solvent. Some solutions are so common to us that we give them a unique name.

e.g. A solution of water and sugar is called syrup. And solution of sodium chloride (common table salt) in water is called brine.

13) Define Unsaturated, saturated and super saturated solutions?

Unsaturated solution:

A solution that is capable of dissolving more solute at a given temperature than it already contains, is known as unsaturated solution.

Saturated solution:

A saturated solution is the solution which can dissolve no more amount of the solute, at a given temperature.

Supersaturated Solution:

A solution that contains more dissolved solute than a saturated solution is called super saturated solution.

14) Write some Properties of Solutions?

1. The particles of solute are the size of individual small molecules or individual small ions. One nanometer is about the maximum diameter for a solute particle.
2. The mixture does not separate on standing. In a gravity environment the solution will not come apart due to any difference in density of the materials in the solution.
3. The mixture does not separate by common filter. The entire solution will pass through the filter.
4. Once it is completely mixed, the mixture is homogeneous. If you take a sample of the solution from any point in the solution, the proportions of the materials will be the same.
5. The mixture appears clear rather than cloudy. It may have some color to it, but it seems to be transparent otherwise. The mixture shows no Tyndall effect. Light is not scattered by the solution. If you shine a light into the solution, the pathway of the light through the solution is not revealed to an observer out of the pathway.
6. The solute is completely dissolved into the solvent up to a point characteristic of the solvent, solute, and temperature. At a saturation point the solvent no longer can dissolve any more of the solute. If there is a saturation point, the point is distinct and characteristic of the type of materials and temperature of the solution.
7. The solution of an ionic material into water will result in an electrolyte solution. The ions of solute will separate in water to permit the solution to carry an electric current.
8. The solution shows an increase in osmotic pressure between it and a reference solution as the amount of solute is increased.
9. The solution shows an increase in boiling point as the amount of solute is increased.

10. The solution shows a decrease in melting point as the amount of solute is increased.

15) Define Solvation, and Dissociation?

Dissociation:

Dissolution means the process of the dissolving or forming a solution. When dissolution happens, the solute separates into ions or molecules, and each ion or molecule is surrounded by molecules of solvent.

Solvation:

The interactions between the solute particles and the solvent molecules are called solvation. A solvated ion or molecule is surrounded by solvent.

16)Write down Properties of Precipitates?

- ☐ ☐ Precipitates are insoluble ionic solid products of a reaction, formed when certain cations and anions combine in an aqueous solution.
- ☐ ☐ The determining factors of the formation of a Precipitate can vary.
- ☐ ☐ Some reactions depend on temperature, such as solutions used for buffers, whereas others are dependent only on solution concentration.
- ☐ ☐ The solids produced in precipitate reactions are crystalline solids, and can be suspended throughout the liquid or fall to the bottom of the solution.
- ☐ ☐ The remaining fluid is called supernatant liquid.
- ☐ ☐ The two components of the mixture (precipitate and supernate) can be separated by various methods, such as filtration, centrifuging, or decanting.

17)Write down Examples and Applications of precipitates?

- ☐ ☐ Precipitation reactions are useful in determining whether a certain element is present in a solution.
- ☐ ☐ If a precipitate is formed when a chemical reacts with lead, for example, the presence of lead in water sources could be tested by adding the chemical and monitoring for precipitate formation.
- ☐ ☐ In addition, precipitation reactions can be used to extract elements, such as magnesium from seawater.
- ☐ ☐ Precipitation reactions even occur in the human body between antibodies and antigens; however, the environment in which this occurs is still being studied.

18)Write down the methods for the measurement of the reaction rate?

1. Continuous Flow
2. Methods for measuring concentration
3. Rate vs. Concentration Proportionalities
4. Relaxation Methods
5. Spectrophotometry
6. Stopped Flow

19)Define Continuous Flow?

Continuous Flow is a type of assay used to easily measure the progress of a reaction at discrete time points and is commonly used for determining initial rates and inhibition values.

- ☐ ☐ In a Continuous Flow experiment, the composition of the reaction is measured continuously, normally by absorbance, while the reactants flow and mix continuously.

□□ This is in contrast to Stopped Flow, where the reactants are mixed in a chamber and then measured, or Batch Reactions, where a number of reactions are performed and stopped at various times by the use of a compound which will halt the reaction.

20) What are Relaxation Methods for the measurement of the reaction rate discuss in detail?

Many chemical reactions are complete in less than a few seconds, which makes the rate of reaction difficult to determine. In these cases, the relaxation methods can be used to determine the rate of the reaction. The term relaxation is used to describe a reaction's return to equilibrium. An equilibrium system is subjected to an external perturbation, such as temperature change. When the change is applied suddenly, the lagging time it takes the system to reach the new equilibrium position is related to the K_f and K_r constants and is called **relaxation time**, τ . There are three different techniques (each are different processes that suddenly disturbs a reaction), that are used to witness the relaxation time.

These are:

- Pressure Jump
- Temperature Jump
- Electric Field Jump

Pressure Jump:

When a reaction is sensitive to changes in pressure, the resulting relaxation can be used to determine the rate constant of reaction. If the reaction is then exposed to a sharp increase in pressure, the reaction will then adjust to a new equilibrium which is controlled by the rate constant(s) for the reaction. Thus, by observing the relaxation of the reaction, the rates can be discovered. This can be utilized for biochemical reactions, such as protein-protein interactions, in order to determine their rate constants.

Temperature Jump:

When there is sudden change in temperature it is called temperature jump. If reaction is in equilibrium and sudden change in temperature happens, the reaction adjusts to the new equilibrium. The change in equilibrium is controlled by the constants of the reaction. The change in rate of the reaction is calculated by observing the shift in the reaction.

Electric Field Jump

The rate of reaction can be calculated by the sudden change in the electric field.

21) Define Spectrophotometry?

Spectrophotometry is a method to measure how much a chemical substance absorbs light by measuring the intensity of light as a beam of light passes through sample solution. The basic principle is that each compound absorbs or transmits light over a certain range of wavelength. This measurement can also be used to measure the amount of a known chemical substance.

22) Define first Order Reactions also give the Differential Representation and Integral Representation?

A first-order reaction is a reaction that proceeds at a rate that depends linearly on only one reactant concentration.

The Differential Representation:

Differential rate laws are generally used to describe what is occurring on a molecular level during a reaction, whereas integrated rate laws are used for determining the reaction order and the value of the rate constant from experimental measurements.

The differential equation describing first order kinetics is given below:

$$\text{Rate} = -\frac{d[A]}{dt} = k[A]^1 = k[A]$$

The "rate" is the reaction rate (in units of molar/time) and k is the reaction rate coefficient (in units of 1/time). However, the units of k vary for non first order reactions.

The Integral Representation:

The integrated first order rate law with respect to a reactant A is:

$$\ln[A] = -kt + \ln[A]_0$$

Where [A] is the concentration at time t and [A]₀ is the concentration at time 0, ln is natural logarithm and k is the first order rate constant. Because the logarithms of numbers do not have any units, the product -kt also lacks units. This concludes that unit of k in a first order of reaction must be time⁻¹. Examples

of time⁻¹ include s⁻¹ or min⁻¹.

23) Describe and verify Mechanism of Chain Reactions?**Mechanism of Chain Reactions:**

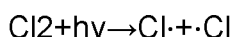
The elementary steps used for mechanisms of chain reactions can be grouped into the following categories:

- Initiation step
- Chain propagation steps
- Chain branching steps
- Chain inhibition steps
- Chain termination steps

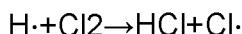
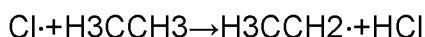
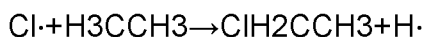
For example, the chlorination of ethane is a chain reaction, and its mechanism is explained in the following way. If we mix chlorine, Cl₂, and ethane, CH₃CH₃, together at room temperature, there is no detectable reaction. However, when the mixture is exposed to light, the reaction suddenly initiates, and explodes. To explain this, the following mechanism is proposed.

Initiation Step:

Light (hν) can often be used to initiate chain reactions since they can generate free radical intermediates via a photodissociation reaction. The initiation step can be written as:

**Chain Propagation Step**

Elementary steps in which the number of free radicals consumed is equal to the number of free radicals generated are called chain propagation steps. Once initiated, the following chain propagation steps repeat indefinitely or until the reactants are exhausted:

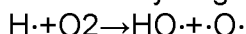


and many other possibilities.

- ☐ ☐ In each of these steps, a radical is consumed, and another radical is generated.
- ☐ ☐ Thus, the chain reactions continue, releasing heat and light.
- ☐ ☐ The heat and light cause more radicals to form.
- ☐ ☐ Thus, the chain propagation steps cause **chain branching reactions**.

Chain Branching Steps

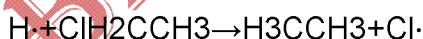
Branching reactions are elementary steps that generate more free radicals than they consume. Branching reactions result in an explosion. For example, in the reaction between hydrogen and oxygen, the following reaction may take place:



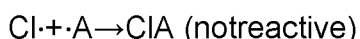
Where $\cdot\text{O}\cdot$ is a di-radical, because the O atom has an electronic configuration $2s^2 2p_x^2 2p_y^2 2p_z^2$. In this elementary step, three radicals are generated, whereas only one is consumed. The di-radical may react with a H_2 molecule to form two radicals. $\cdot\text{O}\cdot + \text{H}_2 \rightarrow \text{HO}\cdot + \text{H}\cdot$. Thus, together chain branching reactions increase the number of chain carriers. Branching reactions contribute to the rapid explosion of hydrogen-oxygen mixtures, especially if the mixtures have proper proportions.

Chain Inhibition Steps

The steps not leading to the formation of products are called inhibition reactions or steps. For example, the following steps are inhibition reactions.



Furthermore, sometimes another reactive substance $\cdot\text{A}$ may be added to the system to reduce the chain carriers to inhibit the chain reactions.

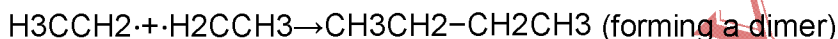
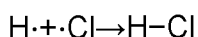
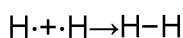
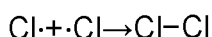


The species $\cdot\text{A}$ is often called a radical scavenger. In food industry, radical scavengers are added to prevent spoilage due to oxidation; these are called biological oxidants. The mechanisms in chain reactions are often quite complicated.

When intermediates are detected, a reasonable mechanism can be proposed. Adding radical scavenger to prevent food spoilage is an important application in food chemistry. This application came from the application of the chain reaction model to natural phenomena.

Chain Termination Steps

Chain termination steps are elementary steps that consume radicals. When reactants are exhausted, free radicals combine with one another to give stable molecules (since unpaired electrons become paired). These elementary steps are responsible for the chain reactions' termination:



24) Define Solution?

A solution is a mixture of materials, one of which is usually a fluid. A fluid is a material that flows, such as a liquid or a gas. The fluid of a solution is usually the solvent. The material other than the solvent is the solute. We say that we dissolve the solute into the solvent. Some solutions are so common to us that we give them a unique name.

e.g. A solution of water and sugar is called syrup. And solution of sodium chloride (common table salt) in water is called brine.